Venkata Lakshmi Parimala Pasupuleti

Software Developer · AI-Enabled Applications · Legacy Modernization · RAG · Guardrails · AWS · Python · Java · Oracle

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SUMMARY

Software developer with 9+ years building backend systems, data integration pipelines, and AI-enabled applications on-prem and on AWS. Currently at Intuit integrating Anthropic Claude into production financial services - RAG pipelines, prompt engineering, structured outputs, AI evaluation, and guardrails. At Entrata (7+ years, 100K+ users), owned mission-critical legacy systems end-to-end: extracted and codified complex federal business rules into data pipelines, led incremental modernization (Docker microservices, AWS S3 migration, async event-driven processing with RabbitMQ), and optimized 200+ Oracle/PostgreSQL queries. Built Leasel - an 8-agent RAG platform with hybrid search, embeddings, guardrails, and a CI evaluation harness (93.6% precision, 1,240 test cases). Strong across the full stack: Python, Java, TypeScript; REST APIs, event-driven architecture, OAuth2/OIDC/JWT; AWS (Lambda, ECS/EKS, API Gateway, SQS, EventBridge, DynamoDB, CloudWatch); Oracle, PostgreSQL, MongoDB. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

AI-Enabled Applications · Foundation Model APIs | RAG Pipelines · Embeddings · Vector Search | Prompt Engineering · Guardrails · AI Evaluation

Legacy Modernization · Business Rules Extraction | Incremental Modernization · Technical Debt Reduction | Backend Development · REST APIs · Microservices

Event-Driven Architecture · Messaging · Async Processing | Enterprise Integration · OAuth2 · OIDC · SAML · JWT

AWS (Lambda · ECS/EKS · API Gateway · SQS · SNS · EventBridge · Step Functions · DynamoDB · CloudWatch) | Oracle · SQL · Query Optimization · Data Modeling

PostgreSQL · MongoDB · DynamoDB · Redis | Python · Java · TypeScript · Node.js · C#/.NET | Docker · Kubernetes · CI/CD · GitHub Actions

Domain-Driven Design · Automated Testing · TDD | Production Monitoring · Observability · OpenTelemetry

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA

- **Integrate Anthropic Claude into production Python/FastAPI financial services** - RAG pipelines for retrieval grounding, prompt engineering for structured outputs, guardrails with input validation and sensitive-data handling, and LLM-as-judge evaluation harnesses with golden datasets to catch hallucinations before release.
- Build ML data pipelines (scikit-learn, XGBoost) from ingestion and feature engineering through prediction APIs; develop React/TypeScript front ends; containerize and deploy services via Docker and GitHub Actions CI/CD to AWS with CloudWatch monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **Legacy modernization and business rules extraction:** Owned the HUD income-limit compliance pipeline for 6 years - analyzed federal source data, extracted and codified complex eligibility and rent-calculation business rules into ~180,000 normalized domain records, with validation, reconciliation, and rollback preparation every release cycle. Zero compliance violations across the full ownership period.
- **Event-driven modernization:** Built RabbitMQ-based async message bus processing for rent updates across thousands of tenant records - retry logic, idempotency guarantees, and failure routing; led Docker microservices redesign (50% scalability improvement, 30% faster deployments); led PHI-compliant AWS S3 migration with Libsodium encryption (100% coverage, zero incidents).
- **Database optimization:** Optimized 200+ Oracle/PostgreSQL SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant schemas for eligibility, financial, and document workflows.
- Delivered 20+ features (Python, Java, PHP, JavaScript, React, PostgreSQL) across 120+ Agile sprints; Python/Bash automation eliminating 80% of manual operational workload; internal debugging tools cutting triage time by 40%; mentored 5+ engineers; 300+ code reviews.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, Oracle/PostgreSQL); delivered large-scale data import and reconciliation workflows (1.6M records); led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - AI Platform: RAG · Agents · Guardrails · Evaluation

2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end AI-enabled application: ingestion pipeline (OCR + document parsing), hybrid search RAG (embeddings, TF-IDF + semantic reranking, citation grounding), 8-agent LangGraph workflow with tool calling and durable state, guardrails enforcing that all regulated decisions route to humans not the LLM, and a CI evaluation harness with LLM-as-judge scoring. 93.6% precision across 1,240 test cases.

AgentProbe - LLM Evaluation & Multi-Provider Routing Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI platform with offline golden-dataset evals, LLM-as-judge scoring, online quality metrics, and a CI regression harness that gates releases on eval thresholds. Cost-aware routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full tracing and structured logging per run.

AI on FHIR — Clinical NLP & Enterprise API Integration 2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates English/Spanish natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut manual clinical lookup time ~60%. HIPAA-style RBAC, OAuth-secured API integration, containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017